Title page

Residual deficits in reactive strength indicate incomplete restoration of athletic qualities following anterior cruciate ligament reconstruction in professional soccer players

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Residual deficits in reactive strength following anterior cruciate ligament reconstruction in soccer players

ABSTRACT

Context: Deficits in plyometric abilities are common following anterior cruciate ligament reconstruction (ACLR). Vertical rebound tasks may provide a targeted evaluation of knee function.

Objective: Examine the utility of a vertical hop test to assess function following ACLR and establish factors associated with performance.

Design: Cross-sectional

Setting: Rehabilitation

Participants: Soccer players with a history of ACLR (N = 73) and matched controls (N = 195)

Main outcome measures: 10 second vertical hop test including measures of jump height, reactive strength index (RSI) and asymmetry. We also examined possible predictors of hop performance including single leg vertical drop jump (SLDVJ), isokinetic knee extension strength, and the international knee documentation committee questionnaire.

Results: Significant between-limb differences were identified for the ACLR group only and asymmetry scores increased in those with a history of ACLR ($P < 0.001$) compared to controls. SLDVJ RSI and knee extension torque were significant predictors of 10 second hop height ($R^2 = 20.1\%$) and RSI ($R^2 = 47.1\%$).

Conclusions: Vertical hop deficits are present following ACLR even after completing a comprehensive rehabilitation program. This may be due to reduced knee extension and reactive strength. Vertical hop tests warrant inclusion as part of return to sport test battery.

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Key words

ACL, Reactive Strength Index, plyometrics, movement assessment, return-to-sport

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INTRODUCTION

Residual deficits following anterior cruciate ligament reconstruction (ACLR) suggest incomplete rehabilitation of the injured limb.[1,2] Vertical jump tests are a valid measure of functional performance and demonstrate only moderate associations with commonly used horizontal hop protocols.[3] Vertical jumping requires greater contributions from the knee to produce the external work performed[4] providing a more targeted evaluation of dynamic knee function. Research indicates a delay in achieving recommended limb symmetry index (LSI) values (> 90%) during vertical jumps compared to horizontal hopping after ACLR.[5] Recent research also showed that single leg drop vertical jumps display greater deficits 9 months after reconstruction compared to the single leg hop for distance in team sport athletes.[6] However, these findings are not consistent across the available research.[7,8]

Measurement of horizontal hop distance and vertical jump height are common when assessing readiness to return to sport.[9] These variables do not account for alterations in jump strategy,[10] lack sensitivity to identify deficits in knee function,[11] and display weak associations with known injury mechanisms including change of direction.[12] When assessing athlete's following ACLR, it is important to consider compensatory strategies. If performing a rebound test on the involved limb, individuals can increase their ground contact time to produce the requisite force needed for propulsion and compensate for reduced stretch shortening cycle function. This athletic ability can be quantified using the reactive strength index (RSI).[13] Longer amortization suggests an inability to attenuate ground reaction forces on landing,[14] and is associated with reduced knee extension force output during the concentric phase of a vertical jump.[15]

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The 10 second single leg hop (10s hop) has been used sparingly to assess functional performance following ACLR.[2,16] This test can account for the limitations in just measuring jump height or hop distance alone. Petschnig et al.[16] reported greater ability to identify performance deficits in ACL reconstructed individuals who were 54 weeks' post-surgery, during the 10s hop test compared to both the single and triple hop for distance. Specifically, LSI scores were below specified thresholds (< 85%) for this functional test only. Similarly, Myer et al.[2] showed significant differences in jump height LSI and normalized vertical ground reaction force at the time of return to sport compared to matched controls. However, neither of these studies included elite athletes and Myer et al.[2] included a force plate. Practically viable assessment modes and normative values are required in elite athletes to help guide practitioners and inform return to sport decision making.

An individual's subjective perception of knee function is also commonly evaluated following ACLR.[17,18,19,20] Previous research has shown moderate correlations between self-reported function and hop performance approximately 6 months' post-surgery.[17,18,19,20] Limited data are available to examine relationships between subjective ratings of knee function and performance during vertical rebound tasks. More symmetrical quadriceps strength has been indicated as an important determinant of safe and effective return to sport,[21] and is associated with greater symmetry during a bilateral drop jump task,[22] but the contribution of quadriceps strength to unilateral rebound tasks is unknown. In addition, RSI is underpinned by stretch-shortening cycle function which is utilized in many sporting movements such as running and change of direction. The most common mode of assessment used to examine this physical attribute has been drop jump protocols using a force platform.[13] Understanding constructs of performance during dynamic tests of knee function using

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practically viable tests and considering both objective and subjective measures can help practitioners design targeted reconditioning programs.

In this study we had two aims; 1) assess functional performance using a practically viable 10s hop test in soccer players who were either currently playing and free from injury or at the end stages of their rehabilitation following ACLR; and 2) examine relationships and predictive factors associated with 10s hop height and RSI to determine the tests performance constructs. We hypothesized that deficits in 10s hop performance would be evident in those with a history of ACLR, and performance on this test would at least in part be explained by knee extension and reactive strength.

METHODS

Experimental Approach to the Problem

ACLR participants and controls attended a formal testing session during which they performed the 10s hop test. Surrogate predictors of 10s hop height and RSI were also examined in ACLR participants only to determine factors associated with successful performance. These included single leg drop vertical jump (SLDVJ) RSI, isokinetic knee extension strength, and patient reported function via the International Knee Documentation Committee (IKDC) questionnaire. All data were collected in a single appointment for ACLR athletes who obtained physiotherapy clearance prior to perform all tests, and for controls as part of a periodic health evaluation. The prescribed test order was as follows: IKDC, SLDVJ, 10s hop, isokinetic assessment.

Participants

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We recruited 268 male soccer players who volunteered to take part in this study. Participants were sub-divided: ***group 1*** (players who were at least 6 months' post ACLR; 24.2 ± 5.1 years; height = 174.9 ± 7.6 cm; mass = 70.9 ± 10.2 kg); and ***group 2*** (non-injured matched controls; 24.7 ± 4.3 years; height = 175.3 ± 6.9 cm; mass = 71.4 ± 9.1 kg). ACL reconstructed players and matched controls were soccer players from teams who compete in one of the recognized leagues as part of the LOCATION-XXX Football Association. Matched controls attended XXX Orthopedic and Sports Medicine Hospital while taking part in an annual periodic health evaluation.

The inclusion criteria for the ACLR group required athletes to be male, having undergone primary surgical reconstruction, and competing as a registered soccer player within one of the recognized competitive leagues of the LOCATION-XXX Football Association prior to their injury. Players were excluded if they reported a history of previous ACL injury / surgery that was not associated with the primary surgical reconstruction; or other knee ligament or cartilage injury / surgery of either the operated or non-operated leg. Matched controls were of the same competitive standard, had no history of ACL or other serious knee injury and were free from severe injury (defined as > 28 days' time-loss) in the previous 12 months, verified by inspection of a national injury audit. Informed written consent and ethical approval was obtained prior to commencement of testing. This study was approved by the Anti-Doping Laboratory, XXX, LOCATION-XXX (IRB: XXX).

Procedures

All players were familiarized prior to testing and completed a standardized warm up consisting of 5 minutes of pulse raising activity and dynamic body weight movements. Only

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1 practice trial was performed of the 10s hop test due to the high intensity and repeated nature of the test protocol. One recorded trial of the 10s hop test was then completed on each limb interspersed with a 60 second rest period. Limb order was randomized. All tests were conducted by an experienced strength & conditioning coach with substantial experience in the relevant test procedures.

10 second single leg hop: Repeated single-leg jumps were performed for a period of 10s in accordance with previous recommendations.[2] The test began with participants placing their hands-on hips and completing a rapid countermovement into a quarter squat followed by a maximal vertical jump. This action was replicated across the test period. Instructions were to jump as high as possible and minimize ground contact time while landing under control, maintain the same footprint, and face forwards during the entirety of the test. Bending of the leg whilst airborne was not permitted. An optical measurement device consisting of a transmitting and receiving bar (Optojump, Microgate, Bolzano, Italy) was used to quantify average jump height and RSI derived from jump height (m) divided by ground contact time (s). LEDs on the transmitting bar communicate continuously with those on the receiving bar and the system detects any disruptions in communication between the bars, recording their duration. This allows measurement of flight and contact time. The first jump was discarded from the analysis as this did not involve a fast stretch-shortening cycle rebound ground contact.

Single leg drop vertical jump: The SLDVJ was performed on a force plate (ForceDecks v1.2.6109, Vald Performance, Albion, Australia) recording at a sampling rate of 1000 Hz. Athletes began in a unilateral stance and dropped off a 15-cm box on their designated test leg. On landing, instructions were to minimize ground contact time and jump as high as possible. Hands were fixed on hips throughout the test and bending of the leg whilst airborne was not

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permitted. Reactive strength index (RSI) was quantified using the equation jump height/ground contact time. Ground contact time was defined as the time from which the vertical ground reaction force exceeded 20 N to the instant of take-off. Jump height was calculated using the athletes centre of mass velocity via the following equation: (COM velocity)^2/ (9.81 x 2). The velocity-time record was obtained by dividing the resultant force-time record by the individuals body mass to provide acceleration, and then numerically integrating with respect to time using the trapezoid rule. Three jumps were performed with the average of all trials reported. This test was selected as the 'gold standard' measure to determine an individual's RSI due to its high frequency of use within the available literature and the development of RSI was originally derived from drop vertical jump protocols.[13] The drop jump is reliable[23] and the task demands athletes must decelerate their downward motion in a yielding / eccentric action until reaching a point of momentary zero velocity and then immediately reverse this motion in a propulsive / concentric action. This assessment provides an effective means of determining stretch shortening cycle function.[13]

Isokinetic knee extension: Maximal knee extension strength was measured using an isokinetic dynamometer (Biodex Medical Systems, Shirley, New York, USA). Players were in a seated position with the hip flexed to 90° with all procedures replicating those of previous research.[24] Assessment modes included five repetitions of concentric knee extension at 60°/s and 10 repetitions at 300°/s with the highest peak torque value recorded across the trials used for analysis, reported as a percentage of the individual's body mass. A 60s rest period was provided between the different testing speeds. Before each test, participants were instructed as to the procedures and provided with appropriate practice repetitions. Vigorous verbal encouragement was provided throughout by the same assessor who conducted all tests.

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International Knee Documentation Committee (IKDC) 2000 questionnaire: The IKDC requires individuals to rate their current level of symptoms, ability to perform specified activities and level of function. The full version of the questionnaire was completed independently by each ACLR participant and scored in accordance with previous guidelines.[25] Players were familiar with the questionnaire and had completed it several times previously during the earlier stages of their rehabilitation. The IKDC questionnaire has been validated, shows high test re-test reliability and is sensitive in identifying lower levels of function.[26]

Statistical Analysis

The distribution of the data was checked using the Shapiro-Wilk normality test. Descriptive statistics (mean ± SD) for all variables were calculated. Between-limb asymmetries in the 10s hop were quantified as a percentage difference between limbs using the following formula: (100/(maximum value)*(minimum value)*-1+100).[27] The maximum and minimum value indicate the highest and lowest performing limb respectively based on the average of all repetitions during the 10s test period. A limb symmetry index which denotes the dominant and operated leg in healthy and ACL-deficient participants, respectively was not used as this presents a challenge when comparing ACLR patients with healthy matched controls because the equation cannot be standardized.

An independent samples t-test was used to examine differences in anthropometrics between groups and paired samples t-test was used to measure between-limb differences for all test variables. Cohen's *d* effect sizes (ES) were calculated to interpret the magnitude of these differences using the following classifications: standardized mean differences of 0.2, 0.5, and 0.8 for small, medium, and large effect sizes, respectively. Performance variables

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(jump height and reactive strength index) and respective asymmetry thresholds were also computed across players in each group by ordering all recorded values and dividing the data for each variable into equal quartiles according to their position with four cut points used (Q1 - Q4).

To determine the constructs measured via 10s hop performance, Pearson's product correlation coefficients were used to examine the strength of the relationships between involved limb 10s average hop height and RSI and the measured surrogate predictor variables including: SLDVJ RSI, isokinetic peak torque measured at 60 and 300°/s (absolute and % BM). Relationships with the IKDC patient reported outcomes was assessed using spearman correlations. The magnitude of these relationships was classified as either; almost perfect (r = >0.9), very large (r = 0.7–0.9), large (r = 0.5–0.7), moderate (r = 0.3–0.5), small (r = 0.1–0.3) or trivial (r = <0.1). Variables that were significantly correlated with either of the dependent variables (10s hop height and RSI respectively) were entered into separate multivariate backwards stepwise linear regression models. Multicollinearity was determined by assessing the variation inflation factor, to ensure values were less than 2.0 in order to eliminate the suggestion of strong multicollinearity with predictor variables included in the model. All data were computed through Microsoft Excel® 2010. t-tests, quartiles, Pearson and spearman correlations, and multivariate linear regressions were processed using SPSS® (V.22. Chicago Illinois). The level of statistical significance was set at alpha level $p \leq 0.05$

RESULTS

No significant differences in anthropometrics were indicated between groups ($P > 0.05$). Descriptive statistics for all variables collected during the 10s hop for ACLR and healthy controls are shown in table 1. Participant groupings by quartiles for 10s hop

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performance and asymmetry scores are shown in table 2. Significant differences were indicated between the involved and un-involved ACLR limbs (P < 0.001; ES = 0.29 - 1.16) but not the dominant and non-dominant limbs of the matched controls for all variables. Between-limb asymmetries ranged from 24.1 – 27.9% and 10.5 – 13.1% for ACLR and matched controls respectively. Significant reductions in asymmetry% were observed in the ACLR group for jump height and RSI.

***** *insert tables 1 and 2 near here* *****

Descriptive statistics for SLDVJ RSI and isokinetic quadriceps peak torque in the ACLR group are shown in table 3. Significant between-limb differences were indicated for all test variables (P < 0.001). Involved limb SLDVJ RSI (r = 0.36), quadriceps knee extension torque @ 60 º/s (% BM) (r = 0.37) and quadriceps knee extension torque @ 300 º/s (% BM) (r = 0.33) showed significant moderate correlations with 10s hop height. Similarly, these variables were significantly correlated with RSI measured during the 10s hop although the strength of these relationships was classified as large for SLDVJ RSI (r = 0.60) and quadriceps knee extension torque @ 60 º/s (% BM) (r = 0.54). Only a moderate relationship was observed for 10s hop RSI and quadriceps knee extension torque @ 300 º/s (% BM) (r = 0.37). Total IKDC score was 77 ± 8 and not significantly correlated related to 10s hop performance.

Prediction models for 10s hop height and RSI are displayed in table 4. Using 10s hop height as the dependent variable, a model including SLDVJ RSI and quadriceps knee extension torque @ 300 º/s (% BM) indicated these variables are significant predictors (P < 0.05) explaining 20.1% of the variance in performance. The prediction accuracy of the model

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improved when 10s hop RSI was used as the dependent variable, with SLDVJ RSI and quadriceps knee extension torque @ 300 °/s (% BM) explaining 47.1% of the variance in performance.

***** *insert table 4 near here* *****

DISCUSSION

The primary aim of this study was to compare the performance of soccer players who had undergone ACLR and were in the final stages of their rehabilitation to those of matched controls in a 10s vertical rebound task used as an indicator of dynamic knee function. Significant differences were identified between the involved and un-involved limbs. Jump height and RSI asymmetry scores were also significantly increased in those with a history of ACLR compared to matched controls. Our secondary aim was to examine relationships and predictive factors associated with 10s hop height and RSI to determine the tests performance constructs. SLDVJ RSI and high velocity knee extension torque were significant predictors of jump height and RSI explaining approximately a quarter and half of the test outcome variance respectively, indicating these physical qualities are contributing factors to vertical rebound tasks emphasising rapid SSC function.

In the current study, significant differences were identified in 10s hop height and RSI between the involved and un-involved ACLR limbs, with asymmetry scores significantly greater compared to matched controls even though they were in the later stages of a comprehensive, evidence-based rehabilitation program.[28] SLDVJ performance in male multidirectional field sport athletes 9 months after reconstruction recently showed significant deficits on the involved leg,[6] with comparative values to those seen in the current study.

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Previously, a 10s vertical jumping protocol identified significant between-limb differences and comparative LSI values (74.9%) for jump height in patients 54 weeks post ACLR.[16] Myer et al.[2] also used the single leg 10s hop tests in a cohort of male and female youth athletes who had returned to sport (9.7 months following reconstruction). They reported significant differences in LSI for the ACLR cohort vs. controls (89% and 101% respectively).

Our findings confirm the that residual deficits in force attenuation and re-utilization capacities are present following ACLR as evidenced by lower 10s hop heights and RSI. Significant between-limb differences shown in SLDVJ RSI with pronounced asymmetries (LSI = ~ 84%) further support this notion. In addition, increased RSI asymmetry in a drop jump has been associated with reductions in change of direction performance.[27] Importantly, change of direction has been recognized as a mechanism of non-contact ACL injury.[29] The results of the current study suggest that targeted interventions are necessary to restore reactive strength. An effective method to increase this physical quality is plyometric training.[14,30] Available evidence indicates that plyometric training is used sparingly during ACL rehabilitation.[31] We encourage practitioners to include this training modality when it is appropriate to do so and propose assessment of RSI using a practically viable tool such as the 10s hop to measure lower limb SSC function. Furthermore, mobile phone applications such as My Jump 2 can also be used, with evidence showing this method provides a valid and reliable tool for the measurement of RSI in vertical jump tasks.[32]

Our results indicate that RSI measured during a SLDVJ and both low and high velocity knee extension strength were most strongly associated with key performance indicators during the 10s hop. Previous research has shown significant relationships between knee extensor strength and vertical jump performance,[33] and slow velocity isokinetic peak torque

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of the quadriceps was moderately associated (r = .51) with 10s hop performance in recreationally trained patients following ACLR.[16] Rate of torque development of the knee extensors is also associated with vertical jump performance.[34] Similarly, knee extension rate of torque development, in addition to RSI measured during a SLDVJ explained 60.9% of the variance in triple hop for distance scores in patients following ACLR.[15] Previous research has reported low velocity knee extension strength to be deficient following ACLR[35] and these residual deficits have been associated with increased risk of future ACL injury.[21] The participants in our study also displayed greater between-limb deficits in quadriceps strength (mean LSI% = 86%; Table 1.) than values previously suggested (LSI > 90%) for safe return to sport.[21] Cumulatively, these results add benefit in patient decision making and indicate that both high and low velocity strength training should also be included as part of a well-structured rehabilitation program to increase lower limb SSC function and enhance an individual's readiness to re-perform. Further research is warranted to examine if lower RSI, and asymmetry in this physical quality is associated with increased risk of secondary injury following return to sport after ACLR.

Intuitively, RSI measured during a SLDVJ and the 10s hop would show some level of correspondence as both are unilateral rebound tasks. A large correlation was reported (r = 0.61) between the two tests but these data indicate a substantial amount of the variance remains unexplained. The SLDVJ is a single rebound task and drop height is constrained by asking participants to drop from a 15cm box. The 10s hop matches landing height to the individual's level of performance capability (i.e. matched to their jump height). In addition, the longer duration and more reactive and repeated nature of the 10s hop test may in part explain why the prediction accuracy of the regression model was increased when RSI was used as the dependent variable. RSI is used to profile stretch load tolerance.[14] Various

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strategies can be employed to maintain jump height (for example longer ground contact times). During repeated jumping tasks whereby, athletes are required to respond to movement perturbations, RSI may provide a more accurate evaluation of an individual's ability to switch from an eccentric to concentric action under varying landing conditions. Thus, the 10s hop including RSI could be considered a meaningful tool to monitor plyometric performance after ACLR. However, further research is required to compare the utility of both the 10s hop and SLDVJ in their ability to identify residual deficits in knee function before confirming its usefulness as a metric to guide readiness to return to sport.

Self-reported measures are frequently included as part of formalized testing to determine an athlete's readiness to return to sport after ACLR. Previous research examining relationships between hop tests and self-reported function has shown variation in which hop tests are associated with self-reported function across the range of studies.[17,18,19,20] In addition, the strength of these relationships is often low to moderate.[17,18,20] In the current study, total IKDC score was not related to 10s hop performance. Self-reported function is multifactorial[20] and these data indicate that the athletes overall patient function score is independent of their ability execute vertical rebound tasks.

An interesting observation from the current study was that mean 10s hop asymmetry scores achieved by the participants for both jump height and RSI were substantially lower than 'pass criteria' (> 90% LSI) recommended during horizontal hop tests for acceptable discharge prior to return to sport.[21] These data are supported by previous research indicating that unilateral vertical jumps display lower LSI scores than horizontal hops at a range of time points (6-24 months) post ACLR.[36] Vertical and horizontal hops could therefore be considered distinctly different tasks by virtue of their moderate associations[3] and measures of

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limb symmetry being task and variable dependent.[37] Furthermore, given the mean asymmetry scores of the control group were also > 10%, and quartile values for players in Q1 and Q2 also exceeded this cut-off, challenging the concept of applying arbitrary thresholds (e.g. > 90% LSI) across all tests and metrics in both performance and rehabilitation environments.

Differences in performance between vertical and horizontal hopping may in part be due to alterations in the lower extremity joint contributions. Chang et al.[34] indicated that knee extensor rate of torque development explained 47% of the variance in vertical jump height. Greater involvement of the hip extensors has been observed when assessing horizontal performance measures[38] and horizontal deceleration of the centre of mass during jump landings.[39] Conversely, the greatest relative total positive work contributions occur at the knee during vertical jumps.[40] Propulsive work contributions of the knee are also lower in horizontal (4%) vs. vertical (24%) jump tasks respectively.[4] Thus, a rebound task requiring vertical acceleration of the body is likely determined more heavily by function of the knee extensors. Due to the residual deficits in quadriceps strength negatively effecting drop jump asymmetry in athletic populations following ACLR,[22] the vertical 10s rebound task may provide an accurate representation of knee joint function and could be used either as an alternative (in cases where time is limited) or addition to more traditional horizontal hopping protocols.

The data presented may be used as normative reference values to guide rehabilitation status and determine an individual's state of readiness to return to sport. Using the descriptive statistics provided in table 1, and in combination with the variables we identified as most associated with 10s hop performance (table 4), clear targets can be established for soccer players. In addition, the quartile values displayed in table 2 provide a simple method, from

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which clinicians can easily identify how their athletes score ranked within a comparable population, and if a player is 'on track' or 'delayed' during their rehabilitation. For example, if you are ~ 9 months' post ACLR (based on mean time post-surgery of the participants in the current study), with a 10s hop height on the involved limb of ≤ 8.2 cm, your score would be in the bottom quartile (Q1), indicative of low performance. Conversely, if a player who is 9 months or more following ACLR shows an LSI score > 82.8%, for RSI, their score would be located in highest quartile (Q4). In this instance, the players score is greater than expected and could be classified as ahead of schedule. In cases where players appear to underperform relative to expected values, further investigation is then warranted to determine possible reasons as to why these elevated residual deficits are present.

When interpreting the results of the current study, it should be considered that our study only measured ACL reconstructed soccer players at a single time point; thus, the temporal recovery of RSI remains unclear. Secondly, although measurement of RSI using an optical measurement system provides an indication of compensatory strategies, no assessment of movement competency was included. Previous research has indicated performance outcomes (hop distance/time) may suggest acceptable symmetry between limbs has been achieved; however, these measures do not consider movement quality which may still remain impaired.[1,6] Recent improvements in wearable technology provide feasible options for clinicians to make more informed and objective decisions. Therefore, other factors relating to neuromuscular control should also be measured and form part of the RTS decision making process. Further research is warranted to more fully examine these approaches, but we feel this area of research provides an exciting development.

PRACTICAL APPLICATIONS

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The results of our study indicate that lower performances and greater asymmetry in 10s hop performances are shown by soccer players who are in the later stages of rehabilitation following ACLR vs. matched controls. These differences may in part be attributed to residual deficits in knee extension and reactive strength as indicated by their predictive associations and the observed between-limb differences in SLDVJ RSI and isokinetic quadriceps peak torque. Therefore, we advise practitioners to train these qualities when appropriate as part of a well-structured reconditioning program when the goal is to return athletes to performance. Finally, greater asymmetry scores shown during the 10s hop test (~ 25%) compared to previous values suggested for acceptable discharge from rehabilitation in traditional horizontal hop tests (e.g. > 90% LSI) may indicate that vertical rebound tasks are more sensitive in their ability to identify deficits in knee function following ACLR.

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Table 1. 10s hop descriptive statistics (mean ± SD)

	ACLR		Healthy controls	
	Involved	**Un-involved**	**Dominant**	**Non-dominant**
Jump height (cm)	10.0 ± 2.7	13.2 ± 2.8*	13.0 ± 3.0	13.2 ± 2.9
Asymmetry (%)	24.1 ± 13.7		10.5 ± 7.8**	
Reactive Strength Index	0.28 ± 0.08	0.39 ± 0.11*	0.41 ± 0.11	0.42 ± 0.10
Asymmetry (%)	27.9 ± 14.5		13.1 ± 9.9**	

* Significant difference between involved and un-involved limb $p < 0.001$
** Significant difference between ACLR and Healthy control groups $p < 0.001$

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Table 2. Asymmetry thresholds for injured and non-injured players in each quartile

	Group	Q1	Q2	Q3	Q4
Jump Height (cm)	ACL-R	< 8.2	8.3 - 9.6	9.7 - 12.3	> 12.4
	Control	< 11.2	11.3 - 13.1	13.2 - 15.1	> 15.2
Asymmetry (%)	ACL-R	> 33.8	33.7 – 24.1	24.0 – 14.1	< 14
	Control	> 16.5	16.4 – 9.9	9.8 – 4.6	< 4.5
Reactive Strength Index	ACL-R	< 0.20	0.21 - 0.27	0.28 - 0.35	> 0.36
	Control	< 0.35	0.36 - 0.40	0.41 - 0.50	> 0.51
Asymmetry (%)	ACL-R	> 39.2	39.1 - 27.6	27.5 – 17.3	< 17.2
	Control	> 18.1	18.0 – 11.1	11.0 – 5.3	< 5.2

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Table 3. Single leg drop vertical jump and isokinetic quadriceps peak torque descriptive statistics (mean ± SD)

Test variable	Involved	un-involved
Single leg drop vertical jump RSI (m/s)	0.79 ± 0.20	0.95 ± 0.21
LSI%	83.7 ± 14.1	
Isokinetic peak torque (% BM) 60°/s	286 ± 47	334 ± 48
LSI%	86.5 ± 13.4	
Isokinetic peak torque (% BM) involved 300°/s	164 ± 28	191 ± 35
LSI%	87.1 ± 14.3	

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Table 4. Linear regression to determine associations with 10s hop performance

Jump variables	Standardized Beta Coefficient	p-value	Adjusted R^2
10s hop height			
SLDVJ RSI	0.358	.007	.201
Knee extension torque @ 300°/s (%BM)	0.321	.015	
10s hop RSI			
SLDVJ RSI	0.593	.000	.471
Knee extension torque @ 300°/s (%BM)	0.369	.001	

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